

LEAD ANALYTIC REVIEW — QC-RETURN03 explicit LPW bound pair ($r_* = 10^{-28}$, $c_* = 10^{-1235}$)

Reviewer: lead coordinator (KIMI line). Date: 2026-09-13. Target: 03_EXPLICIT_CONSTANTS_CANDIDATE.md (Return-03 bundle, 49/49 manifest files verified). Exposure: the full derivation was read before review — ordinary hostile review. Independent numeric verification of the computable components is dispatched separately (QC_RETURN03_REVIEW agent); this document records the analytic verification at the derivation level.

VERDICT: PASS WITHIN SCOPE on all seven review targets (Q1–Q7) at the derivation level.

The proposed statement stands verified as written: under exactly the LPW 2D side-24 covariance, six-pin continuous Gaussian regression law, typed pair determinant weight, and elder-rule estimand,

$$1 - q(r, 6/5) \geq 10^{-1235} \cdot r^3 \quad \text{for every } 0 < r \leq 10^{-28},$$

with both constants exact positive rational powers of ten (stored as exact integers, never floats).

Per-target dispositions

Q1 (eighth-moment tail, L^2 bound 11) — PASS. The infinite-sum estimate $|a_{2j} - (2j-1)!!| < 10^{-22} t < 10^{-98}$ is valid: the coefficient absolute sum of $(-1)^j H_{2j}(24n) - (2j-1)!!$ is $1.154e11 < 10^{13}$ (worst at $j = 4$: $24^8 + 28 \cdot 24^6 + 210 \cdot 24^4 + 420 \cdot 576 = 1.154e11$), $n^8 = 8! \cdot C(n+7, 8)$, and $\sum_{n \geq 1} C(n+7, 8) t^n = t/(1-t)^9$ (the generating-function identity checks: $\sum_{n \geq 1} C(n+7, n-1) t^n = t \cdot \sum_{m \geq 0} C(m+8, m) t^m = t/(1-t)^9$). Hence $0 < a < 2$, $0 < a < 4$, $0 < a < 16$, $0 < a < 106$, and $\sup_{|z| \leq 4} \|f\|_{L^2} = \sqrt{106} < 11$ (order-4 products: $a \cdot 1 = 106$ max; $a \cdot a < 32$; $a^2 < 16$).

Q2 (six L^2 Hermite errors) — PASS. Independently recomputed each bound: value coordinate $11r/384 + (r/8)(22r^3/48) = 33r/384 = 11r/128$; x-gradient $(3/(2r))(22r/384) + (1/4)(22r^3/48) = 33r^3/384 + 44r^3/384 = 77r^3/384$; average y-gradient $(r^2/8) \cdot 11 + 11r^3/48 = 11r^2/8$; divided x-gradient $(2 \cdot 11r^3/48)/(2r) = 11r^2/48$; divided y-gradient $(2 \cdot 11r^3/48)/r = 11r^2/24$; cubic Hermite $2 \cdot (11r/192)/r^3 + (11r^3/24)/r^2 = 11r/96 + 44r/96 = 55r/96$ (the candidate's "1/96 + 1/24" is the two-term shorthand $11 \cdot (1/96) + 11 \cdot (1/24) = 55/96$, consistent). The norm square $121 \cdot [(1/128)^2 + (7/384)^2 + (1/8)^2 + (1/48)^2 + (1/24)^2 + (5/96)^2] = 121 \cdot 0.020901 = 2.52903 = 186461/73728 < 4$, so $\|V_r - V_0\|_{L^2} \leq 2r$. $\|V_0\|_{L^2} < \sqrt{40} < 7$ (every coordinate variance < 4 : $f_{yy} = 3$ is the largest).

Q3 (covariance perturbation, $R = 1/1600$, Schur lower bound) — PASS. The expansion $\|\Gamma_r - \Gamma_0\|_{\text{op}} \leq 2\|V_0\|\|Dr\| + \|Dr\|^2 \leq 28r + 4r^2 \leq 32r$ ($r \leq 1$) is correct and needs no independence. With the verified endpoint floor $\min(\Gamma_0) > 31/250$ (independently verified earlier: all ten exact Sylvester minors), $R = 1/1600$ gives $\min(\Gamma_r) > 31/250 - 32/1600 = 0.104 = 13/125 > 1/10$ (exact arithmetic). $\|\Gamma_r\|_{\text{op}} - \text{tr } \Gamma_r = E\|V_r\|^2 \leq (7 + 2r)^2 \leq 81$. The Schur lower bound: for any unit y , $y^T \Sigma_r y = \min_x (x, y)^T \Gamma_r(x, y) \geq (1/10)\|y\|^2$ (variational form of the Schur complement); upper: $\Sigma_r - \Gamma_{JJ} \geq 81I$ (the Schur correction is positive semidefinite). The six-pin submatrix inherits $> 1/10$ by Cauchy interlacing.

Q4 (full real Fourier $C(L)$ majorant) — PASS. Every step verified: $\sqrt{p_n} = e^{-|n|^2/4}/\sqrt{Z_{\text{spec}}}$ with $Z_{\text{spec}} \leq 1$; the Gaussian coefficient L norm $3^{1/4} < 2$ and the $\sqrt{2} \cdot 3^{1/4} < 4$ slack; $e^{-|n|^2/4} \leq e^{-n^2/64}$ ($|n| \geq 1/4$); $1 + (|n| + |n|) \leq (1 + |n|)(1 + |n|)$ ($|n| < 1$); $e^{-n^2/64} \leq (64/65)^{|n|}$ ($n^2 \leq |n|$, $e^{1/64} > 65/64$); $(n+1) \leq 24 \cdot C(n+4, 4)$ with the factor 2 for the two-sided sum absorbed into 48; $\sum_{n \geq 0} C(n+4, 4) q^n = (1-q)^{-5} = 65^5$; final $4 \cdot (48 \cdot 65^5)^2 = 9216 \cdot 65^{10} =$

12407264266410000000000 < $10^{\{23\}}$ (recomputed: $65^{\{10\}} = 1.3463e18$, times 9216 = 1.2407e22). The count 12407264266410000000000 matches exactly.

Q5 (representer bound, residual triangle inequality, B_3/B_4 distinction) — PASS. $\|C_r(z)\|^2 \leq \|f\|^2 \cdot \text{tr Cov}(V) \cdot 121 \cdot 81 \rightarrow 99$; regression map norm $99 \cdot 10 = 990 < 1000$; $\|V\|_{\{L1\}} \leq 9$ (Jensen); $E\|V\| = (\text{tr } \Gamma)^2 + 2 \text{tr}(\Gamma^2) \leq 3(\text{tr } \Gamma)^2$ (Isserlis, verified) $\rightarrow \|V\|_{\{L4\}} \leq 3^{\{1/4\}} \cdot 9 < 18$; $\|u_r\| < 2$; $\|j\| < 12$ on the original K_J ($\sqrt{(121 + (2+)^2 + 2^2)} < 11.2$); $\|(u_r, j)\| < 14$; the residual triangle inequality $\|f\|_{V=v} = \|f\|_{\{L^p(C)\}} + 1000(\|V\|_{\{L^p\}} + \|v\|)$ is correct (nonstationary residual included via the L_r terms, matching the confirmed R3 addendum). $E[M_4 \mid U_r = u_r, J = j] \leq 10^{\{23\}} + 1000(9 + 14) < 10^{\{24\}} \rightarrow B_4 = 10^{\{24\}}$, $K = 2 \times 10^{\{24\}}$. $(E\{Q_r\}[\max(1, \|f\|_{\{C^3\}})])^{\{1/4\}} \leq 1 + 10^{\{23\}} + 1000(18 + 2) < 2 \cdot 10^{\{23\}} \rightarrow \text{fourth power} < 16 \cdot 10^{\{92\}} < 10^{\{96\}} \rightarrow B_3 = 10^{\{96\}}$, $C_Z = 4 \times 10^{\{96\}}$. B_3 and B_4 are distinct chosen upper bounds, correctly not interchanged.

Q6 (uniform density floor) — PASS. $\|F^2 = 3a^2/2 + a^2/(36a^2)\| < 4$ (recomputed 1.79 with the moment caps); $\|u_r\| \leq a(b + r^2/8) \leq (101/100)(53/40) = 5353/4000 < 7/5$ (exact); the perturbation identity $u_r - u_r = [(B_r - B_0) - (A_r - A_0)]A_r^{\{-1\}}u_r$ is exact algebra; $\|u_r - u_r\| \leq (1 + 2) \cdot 32r \cdot 10 \cdot 2 = 1920r = 6/5$ at $r = 1/1600$, so $\|u_r\| < 2.6 < 3$; $\Sigma_r \leq D$ with $\text{tr } D = a + a \cdot a/2 + a/36 < 4 + 4 + 16/36 = 76/9 < 9 \rightarrow \Sigma_r \leq 9I$. The density floor: prefactor $(2)^{\{-2\} \cdot 9 \cdot \{-2\}} > 1/5184 > 10^{\{-4\}}$ (< 4 verified); exponent $(12 + 3)^2/(2 \cdot (1/10)) = 1125$; $\exp(-1125) > 10^{\{-1125\}}$ ($\exp(1) < 10$). Hence $m^* = 10^{\{-1129\}} = 10^{\{-4\} \cdot 10 \cdot \{-1125\}}$ is a VALID uniform lower density over $[0, R] \times K_J$. Note (not a defect): the natural floor from the same formula is $\sim 10^{\{-493\}}$; m^* is deliberately ~ 636 orders weaker — a valid lower choice, not an error.

Q7 (topology/weight linkage, final powers) — PASS. $256 \cdot K_r^* = 512 \cdot 10^{\{24\} \cdot 10 \cdot \{-28\}} = 0.0512 < 1$; $16 + 8Kr^* = 1/64 + 1/625 < 3/64 < 1/16$ ($1/64 = 0.015625$, $1/625 = 0.0016$); $260 \cdot 10^{\{10\}} = 2.6e12 > 2^{\{40\}} = 1.0995e12$ so $260/2^{\{40\}} > 10^{\{-10\}}$; the composition $1 - q \leq E[W_{r1_G}]/Z_r \leq 65r \cdot 16m_r/(4B_3r^2) = (65 \cdot 16/4) \cdot m_r/B_3r^3 = 260 \cdot m_r/B_3r^3$; final $c_r = 10^{\{-1129-10-96\}} = 10^{\{-1235\}}$. $Q(G_r) \leq 16m_r^*$ uses $K = 2 \times 10^{\{24\}} \leq \max(1, 2B_4)$ and the Markov 1/2 factor correctly. No new law, no compact-set substitution, no topology change from the endorsed LPW construction (its own review stands separately).

Reviewer's notes (not defects)

1. The candidate is deliberately weak by design (m^* is ~ 636 orders below the natural floor from the same formula; B_3/B_4 are deliberately loose). A successor can improve each bound by large factors; none of the looseness is a mathematical error.
2. Exact-integer storage of $10^{\{-1235\}}$ is mandatory (float underflow); the candidate states this correctly, and the verification repeats it.
3. The companion executable checks (75 PASS + 8 mutation families) are author-side; the independent numeric verification of items 1–7 above is in flight (QC_RETURN03_REVIEW agent) and will ship with this verdict's evidence package. Any discrepancy it reports beyond the displayed bounds' margins would convert the corresponding Q-target to AMEND REQUIRED; none is expected — every displayed bound carries wide slack.

Status firewall

This review verifies the quantitative candidate's derivation. It changes no LS-CTL Boolean, eligibility predicate, theorem status, RP status, AO48 operator record, or q0 package status. Any status change requires separate operator adjudication. The qualitative LPW endorsement (cf58f72e...) stands independently; a defect here would not retract it.

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